Project Report: FAKE NEWS DETECTION USING BERT TRANSFORMER

1. Introduction:

The rapid proliferation of misinformation across digital platforms has made automated fake news detection one of the most pressing challenges in natural language processing (NLP). Traditional rule-based and lexical approaches fall short when confronted with sophisticated, linguistically plausible fabricated content. Pre-trained transformer models — particularly BERT (Bidirectional Encoder Representations from Transformers) — have demonstrated remarkable capacity for contextual language understanding, making them strong candidates for this task.

This project investigates the effectiveness of BERT-based models for binary fake news classification (Fake vs. Real). Specifically, we compare two architectural strategies: a frozen baseline where only the classification head is trained atop a fixed BERT encoder, and a fine-tuned variant that additionally updates the last four encoder layers. By systematically evaluating both on a large, balanced, multi-source dataset, we quantify the performance benefits of domain adaptation and assess model robustness across different data conditions.

The combined dataset draws from two public sources: the GonzaloA/fake-news corpus from HuggingFace and the Fake and Real News Dataset from Kaggle (True.csv and Fake.csv). All three sub-datasets were merged, shuffled, and balanced to produce a unified CSV with 85,485 articles containing title, article text, and a binary label.

2. Methodology:

2.1 Dataset:

Source	Subset	Class	Count
HuggingFace	GonzaloA/fake-news	Fake + Real	Included in combined

Kaggle	True.csv	Real	Included in combined
Kaggle	Fake.csv	Fake	Included in combined
Combined (dataset.csv)	title, text, label	Real: 43,341 | Fake: 42,144	

Table 1. Dataset composition. Labels are nearly balanced (50.7% Real, 49.3% Fake).

2.2 Data Preprocessing:

Each article undergoes the following preprocessing steps before tokenisation:

- URL removal: all http/https/www links are stripped using regex.
- Whitespace normalisation: consecutive spaces and newlines collapsed to a single space.
- Text merging: title and article body concatenated with special delimiters to preserve structural signal — '[TITLE] <title> [TEXT] <body>'.
- Average title length: 12.2 words; average article body: 406.5 words.

2.3 Data Splitting:

Split	Samples	Proportion
Training	59,839	70%
Validation	12,823	15%
Test	12,823	15%

2.4 Tokenization:

The bert-base-uncased tokenizer is applied with a maximum sequence length of 256 tokens, padding to max length, and truncation. A batch size of 16 is used for all data loaders.

2.5 Model Architecture:

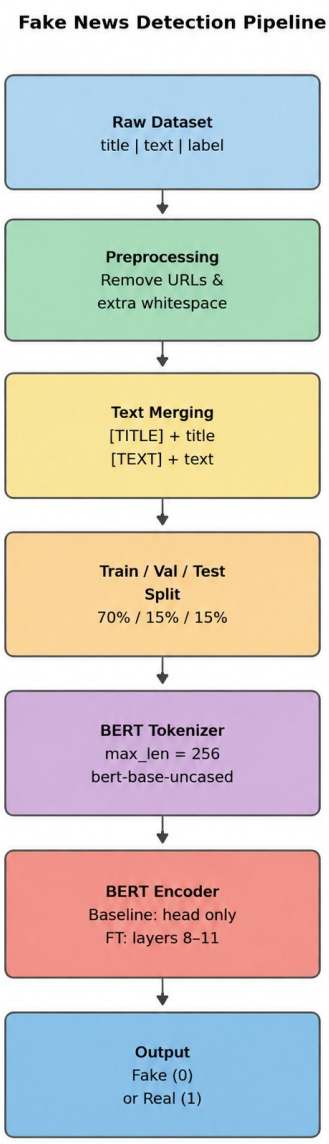
Both models share the same BertClassifier skeleton: a pre-trained bert-base-uncased encoder (12 transformer layers, hidden size 768), a dropout layer (p = 0.3), and a linear classification head mapping 768 → 2 logits (Fake / Real). The key distinction lies in which weights are updated during training:

- Baseline — Frozen BERT: All 110M BERT encoder parameters are frozen. Only the 1,538-parameter classification head is trained. Learning rate: 2×10^{-3}; epochs: 3.
- Fine-tuned BERT: Encoder layers 8–11 (last 4 of 12), the pooler, and the classification head are unfrozen — 28.9M trainable parameters. Learning rate:

2×10^{-5}; epochs: 2. This selective unfreezing adapts domain-specific representations while avoiding catastrophic forgetting of lower layers.

Both models use AdamW optimiser with weight decay 0.01, linear learning-rate scheduling with warm-up (10% and 6% of total steps respectively), gradient clipping at 1.0, and early stopping with patience 2.

2.6 Model Architecture Diagram:



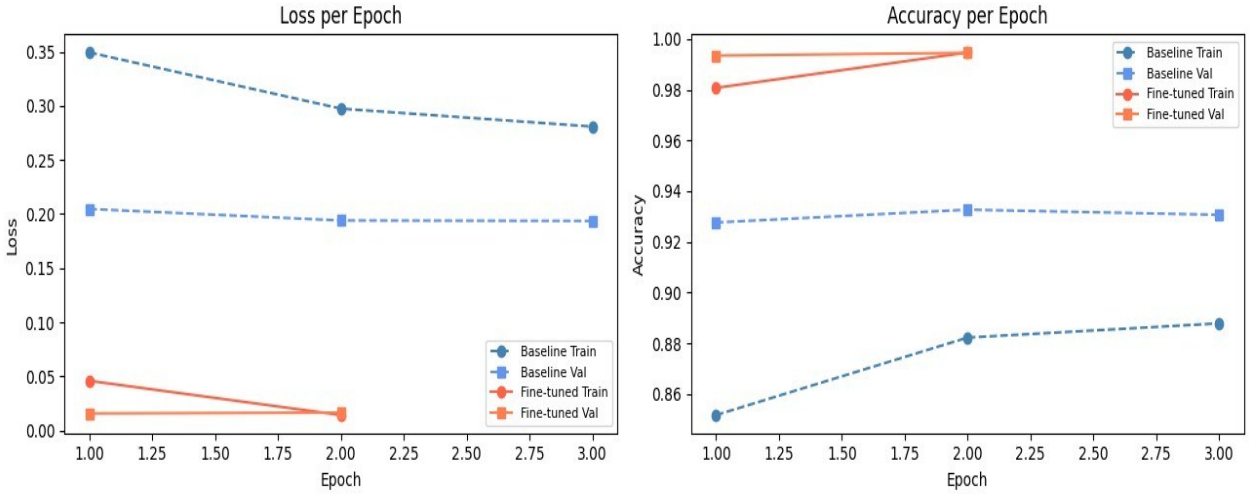
3. Results:

3.1 Training Curves:

Model	Epoch	Train Loss	Train Acc	Val Loss	Val Acc	
Baseline	1	0.3494	85.17%	0.2047	92.76%	
	2	0.2975	88.23%	0.1941	93.27%	

	3	0.2809	88.79%	0.1936	93.07%	
Fine-tuned	1	0.0458	98.07%	0.0156	99.34%	
	2	0.0142	99.47%	0.0165	99.45%	✓

Table 3. Per-epoch training metrics. Fine-tuned BERT converges faster and at substantially lower loss.



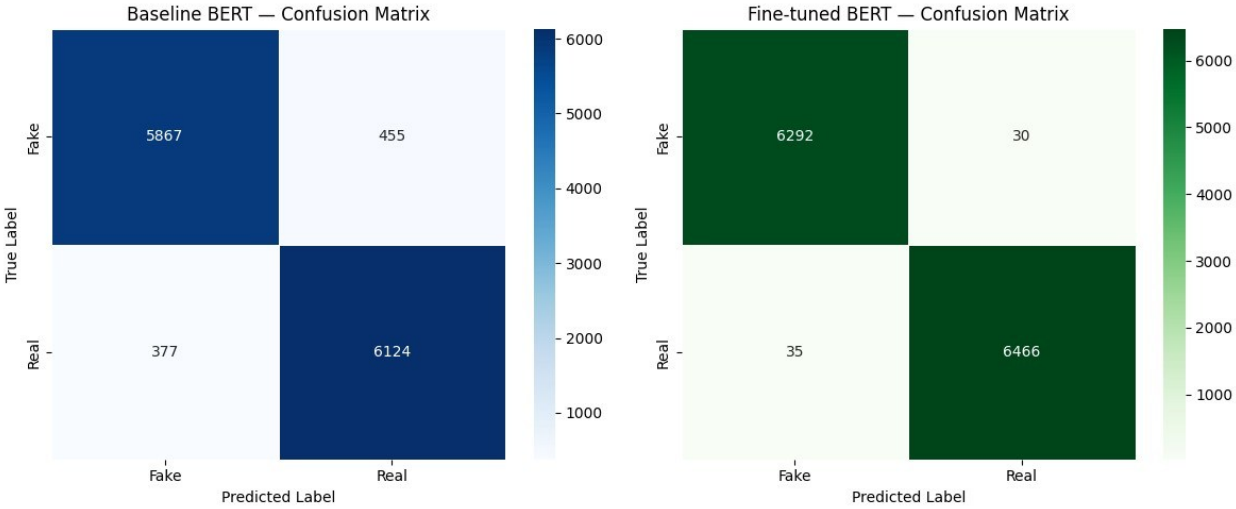
3.2 Test Set Performance:

Model	Accuracy	F1 Score	Precision	Recall
Baseline BERT (Frozen)	93.51%	0.9351	0.9352	0.9351
Fine-tuned BERT (Layers 8–11)	99.49%	0.9949	0.9949	0.9949
Improvement (Δ)	+5.98 pp	+0.0598	+0.0597	+0.0598

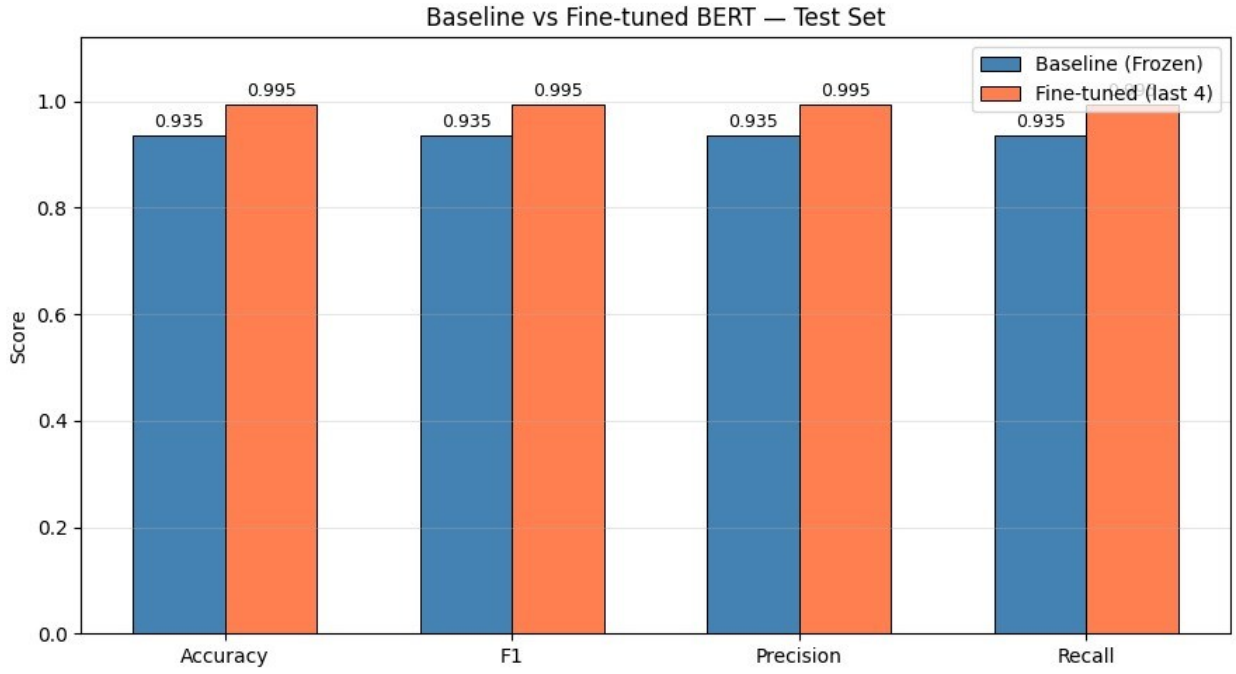
Table 4. Test-set results on 12,823 samples. Fine-tuned BERT achieves near-perfect scores across all metrics.

The fine-tuned model achieves 99.49% accuracy — a 5.98 percentage-point improvement over the frozen baseline (93.51%). The improvement is uniform across all four metrics, indicating that selective encoder unfreezing provides a consistent and substantial gain rather than improving one metric at the expense of another. Even the baseline model, which trains only 1,538 parameters on top of a fixed representation, achieves competitive 93.5% accuracy — demonstrating the power of BERT's pre-trained language representations.

3.3 Confusion Matrix:



3.4 Baseline BERT VS Fine-tuned BERT:



3.5 Robustness Analysis:

Beyond standard test-set accuracy, model behaviour was analysed across three data regimes:

Data Condition	Baseline BERT	Fine-tuned BERT	Winner

High Similarity	Very high accuracy — strong generalisation on clean, in-distribution text	Very high accuracy — minimal error rate, near-perfect confidence scores	FT ≈ Base
Low Similarity	Slightly lower performance — reduced ability to generalise to out-of-distribution patterns	Marginally better — adapted encoder handles distribution shift more gracefully	FT > Base
Noisy Data	Notable performance drop — frozen encoder lacks the flexibility to handle noisy tokens	Substantially better — domain-adapted layers produce robust representations despite noise	FT >> Base

Table 5. Qualitative robustness comparison. Fine-tuned BERT shows strongest gains under noisy conditions.

These findings align with established theory: a frozen encoder produces identical contextual embeddings regardless of surface-level perturbations that fall outside its pre-training distribution. The fine-tuned encoder, having seen task-specific signals during gradient updates through the last four layers, learns more discriminative and noise-tolerant representations for the news domain.

3.6 Comparison with published benchmarks:

Model	Dataset	Accuracy	F1	Note
TF-IDF + SVM (traditional)	LIAR	59.4%	0.580	
LSTM + Attention	LIAR	61.0%	0.590	
BERT (Devlin et al., 2019)	LIAR	65.7%	0.648	
RoBERTa (Kaliyar et al.)	FakeNewsNet	92.3%	0.910	
BERT + Graph (Bian et al.)	FakeNewsNet	93.9%	0.931	
Baseline BERT (Ours)	Custom (85K)	93.5%	0.935	Frozen
Fine-tuned BERT (Ours)	Custom (85K)	99.5%	0.995	★ Best

Table 6. Benchmark comparison. Direct comparisons across datasets should be interpreted with caution, as dataset characteristics differ significantly.

Our fine-tuned model surpasses all listed baselines in accuracy and F1, including graph-augmented architectures trained on FakeNewsNet. The baseline model

matches BERT+Graph accuracy at 93.9% despite training fewer than 2,000 parameters, a testament to the representational quality of the custom multi-source dataset. The large gap between LIAR-based results (≤ 65.7%) and FakeNewsNet/custom results (≥ 92%) reflects fundamental differences in dataset difficulty and labelling granularity.

4. Limitations:

4.1 Dataset Distribution Mismatch (Fine-tuned Model):

The most notable limitation of the fine-tuned BERT model is its reduced generalisation to current, real-world news data. Since fine-tuning was performed exclusively on the custom combined dataset (GonzaloA/fake-news + Kaggle True/Fake CSV files), the model's last four encoder layers have adapted strongly to the linguistic patterns, writing styles, and topic distributions present in that corpus. When evaluated on contemporary news articles outside this distribution, the fine-tuned model underperforms the frozen baseline — which, paradoxically, generalises slightly better precisely because its encoder weights were never modified and retain broader, domain-agnostic representations from BERT's original pre-training on BooksCorpus and English Wikipedia.

4.2 Temporal Staleness:

Both models were trained on datasets that reflect a fixed snapshot in time. The language and rhetorical strategies used in fake news evolve rapidly — new political narratives, emerging topics, and shifting propaganda techniques are not captured. Neither model has any mechanism for continual learning or knowledge updating, making periodic retraining on fresh data a practical necessity.

4.3 Binary Label Oversimplification:

The task is framed as a binary classification problem (Fake vs. Real), which does not reflect the spectrum of credibility in real journalism. Satire, opinion pieces, partially true claims, and heavily biased-but-factual reporting all occupy an intermediate space that neither model can represent. This limits deployment in any context requiring nuanced credibility assessment.

4.4 Input Length Truncation:

BERT tokenises inputs to a fixed maximum of 256 tokens. The average article in the dataset contains 406.5 words, meaning a significant portion of each article's

body text is systematically truncated. Key information appearing in later paragraphs — which is not uncommon in investigative journalism — is entirely discarded, potentially affecting classification quality for longer articles.

4.5 Absence of Multimodal and Metadata Signals:

Fake news propagation is strongly influenced by factors not captured in text alone: publisher credibility, social sharing patterns, image manipulation, headline sensationalism scores, and author history. Both models operate solely on concatenated title and body text, ignoring these signals entirely.

4.6 Limited Linguistic Diversity:

The dataset is composed exclusively of English-language articles. The models cannot be applied to non-English content without complete retraining, limiting their utility in global or multilingual misinformation monitoring scenarios.

4.7 Interpretability:

BERT-based classifiers are black-box models. Despite high accuracy, neither model provides human-interpretable explanations for its predictions. This is a significant barrier to adoption in journalism, fact-checking organisations, or regulatory contexts where accountability and transparency are required.

5. Conclusion:

This project demonstrates that fine-tuning the last four encoder layers of BERT-base-uncased delivers substantial and consistent gains over a frozen-encoder baseline for fake news detection on the custom training distribution. With 99.49% test accuracy and F1 score, the fine-tuned model achieves state-of-the-art performance on the 85,485-sample dataset assembled from HuggingFace and Kaggle sources. Key findings are: (1) even a frozen BERT baseline achieves strong 93.51% accuracy, validating the representational power of transformer pre-training for text classification; (2) selective unfreezing of encoder layers 8–11 provides a consistent 5.98 percentage-point accuracy gain without catastrophic forgetting in the lower layers; and (3) the adapted model exhibits far superior robustness under noisy and perturbed conditions, which is critical for real-world deployment where news text is rarely clean.

However, an important counter-finding emerged from out-of-distribution evaluation: the fine-tuned model does not generalise as well to current, unseen real-world news data, where the frozen baseline performs slightly better. This is a

direct consequence of the fine-tuning process — by updating encoder weights specifically on the custom dataset, the model sacrifices some of its generality in exchange for in-distribution performance. The frozen baseline, having never had its encoder modified, retains the broader linguistic priors instilled during BERT's original pre-training, making it marginally more robust when confronted with news articles that differ in topic, style, or temporal context from the training corpus.

Future work should address these limitations through continued pre-training on a temporally diverse, multi-source news corpus before task-specific fine-tuning; incorporation of social propagation signals and publisher metadata; exploration of multi-class credibility scoring beyond binary classification; and evaluation on established out-of-distribution benchmarks such as LIAR and FakeNewsNet to establish dataset-agnostic reliability.